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Define "immediate context"

This paper aims to provide the definition for the term "immediate context". This addresses [US54-100](#).

1. EWG guidance

The following three polls have been taken on 2026-03-24 during the Croydon meeting.

Poll1: As an answer to US 54-100 for CWG, default arguments ARE part of the immediate context

SF	F	N	A	SA
0	2	12	10	6

Verdict: not consensus.

Poll2: As an answer to US 54-100 for CWG, default arguments ARE NOT part of the immediate context

SF	F	N	A	SA
5	12	11	2	1

Verdict: consensus.

Poll3: Opinion: default member initializers of aggregates should be part of the immediate context. Consensus may be called either for/against.

SF	F	N	A	SA
1	1	12	10	3

Verdict: Initially incorrectly declared "no consensus", then corrected to "consensus against".

2. Further questions for EWG

CWG would like to unify the treatment of noexcept-specifiers, function-contract-specifiers, default arguments, and annotations in order to improve the consistency of the language and the specification. The proposed unified treatment is that the immediate context excludes separately instantiated constructs and includes everything else (other than lambda bodies). To that end, CWG is requesting EWG's consent for the following changes/clarifications. We believe that items 1 and 2 below apply only to the special case of constructs that appear in a non-generic lambda in a SFINAE context (like the return type of a function template).

2.1. Item 1

Default arguments are always subject to separate instantiation, even when they appertain to non-generic lambdas or member functions of local classes. However, in non-generic lambdas and member functions of local classes, that separate instantiation is not deferred; consequently, default arguments are excluded from the immediate context even in these cases. (For local classes, this is not a change in behavior. For non-generic lambdas, the status quo is unclear). For example:

```
struct X {};  
  
// default argument not used:  
template <class T>  
auto foo() -> decltype([](T x = T(1)){T()});  
  
template <class T>  
concept canFoo = requires { foo<T>(); };  
  
constexpr bool b = canFoo<X>; // ???
```

CWG suggestion: There is no separate instantiation for the default argument in this case. A hard error therefore occurs during overload resolution, even though the default argument isn't used.

Implementation status: GCC currently accepts; Clang and MSVC reject.

2.2. Item 2

noexcept-specifiers on function declarations will be changed so that they are treated the same way as suggested above for default arguments, i.e., to receive separate instantiation always (and thus never be in the immediate context) but have that separate instantiation be performed immediately in the non-generic lambda and local class cases. (The status quo is that *noexcept-specifiers* of non-generic lambdas and member functions of local classes are not subject to separate instantiation. In the lambda case, Clang and GCC treat them as being in the immediate context.) For example:

```
template <class T>  
auto foo() -> decltype([]() noexcept(noexcept(T() + 1)){ /* ... */ });  
  
template <class T>  
concept canFoo = requires { foo<T>(); };  
  
struct X {};  
  
constexpr bool b = canFoo<X>; // ???
```

Status quo: `b` is initialized to `false`.

Implementation status: Clang and GCC agree, while MSVC gives a hard error.

CWG suggestion: the result should be a hard error because the *noexcept-specifier* is immediately but separately instantiated.

If this suggestion is rejected, then *noexcept-specifiers* will be self-consistent (not separately instantiated in the above context + yes in the immediate context) but different from function contract specifiers.

2.3. Item 3

Annotations of templated entities should be instantiated only when needed, similarly to *noexcept-specifiers* and function contract specifiers, and are therefore never in the immediate context. (CWG has not yet formulated a position as to when, exactly, annotations are considered needed.) Unlike in items 1 and 2, this clarification would affect templated functions in general, not just the lambda edge cases.

3. Wording

The proposed wording is relative to [\[N5032\]](#).

Edit [temp.decls.general]/3:

For purposes of name lookup and instantiation, default arguments, type-constraints, requires-clauses ([temp.pre]), ~~and~~ *noexcept-specifiers*, ~~and~~ *function-contract-specifiers* of ~~function templates and of member functions of class templates~~ *templated functions* are considered definitions; each default argument, type-constraint, requires-clause, ~~or~~ *noexcept-specifier*, ~~or~~ *function-contract-specifier* is a separate definition which is unrelated to the templated function definition or to any other default arguments, type-constraints, requires-clauses, or *noexcept-specifiers*. For the purpose of instantiation, the substatements of a *constexpr* if statement are considered definitions. For the purpose of name lookup and instantiation, the compound-statement of an expansion-statement is considered a template definition. For the purpose of instantiation, an *annotation* is considered a definition.

Strike note 3 from [temp.inst]/2:

~~{Note 3: Within a template declaration, a local class or enumeration and the members of a local class are never considered to be entities that can be separately instantiated (this includes their default arguments, noexcept-specifiers, and non-static data member initializers, if any, but not their type-constraints or requires-clauses). As a result, the dependent names are looked up, the semantic constraints are checked, and any templates used are instantiated as part of the instantiation of the entity within which the local class or enumeration is declared. — end note}~~

Edit [temp.inst]/3:

~~[...] The implicit instantiation of a class template specialization does not cause the implicit instantiation of default arguments or noexcept-specifiers of the class member functions. [...]~~

Edit [temp.inst]/5:

Unless a function template specialization is a declared specialization, the function template specialization is implicitly instantiated when the specialization is referenced in a context that requires a function definition to exist or if the existence of the definition affects the semantics of the program. A function whose declaration was instantiated from a friend function definition is implicitly instantiated when it is referenced in a context that requires a function definition to

exist or if the existence of the definition affects the semantics of the program. ~~Unless a call is to a function template explicit specialization or to a member function of an explicitly specialized class template, a default argument for a function template or a member function of a class template is implicitly instantiated when the function is called in a context that requires the value of the default argument.~~

Add a paragraph before [temp.inst]/12:

The default arguments, *noexcept-specifier*, and *function-contract-specifiers* of a templated function that is a member of a local class or the function call operator for the closure type of a non-generic *lambda-expression* are instantiated when the declaration of the function is instantiated. [Note: For the purposes of instantiation, these constructs are always considered separately from the function to which they belong. ([temp.decls.general]). — end note]

Edit [temp.inst]/12:

Other than as specified above, when^{if} a templated function *f* is called in a way that requires a default argument to be used, the dependent names are looked up, the semantics constraints are checked, and the instantiation of any template used in the default argument is done as if the default argument had been an initializer used in a function template specialization with the same scope, the same template parameters and the same access as that of the function template *f* used at that point, except that the scope in which a closure type is declared ([expr.prim.lambda.closure]) — and therefore its associated namespaces — remain as determined from the context of the definition for the default argument. This analysis is called default argument instantiation. The instantiated default argument is then used as the argument of *f*.

Edit [temp.inst]/14:

Other than as specified above, the^{The} *noexcept-specifier* and *function-contract-specifiers* of a ~~function template specialization of a templated function~~ ^{are not instantiated along with the function declaration; they are} instantiated only when needed ([except.spec], [dcl.contract.func]). ^{if}When such a specifier is ~~needed but has not yet been~~ instantiated, the dependent names are looked up, the semantics constraints are checked, and the instantiation of any template used in the specifier is done as if it were being done as part of instantiating the declaration of the specialization at that point.

Edit [temp.inst]/17:

The type-constraints and *requires*-clause of a template specialization or member function are not instantiated along with the specialization or function itself, ~~even for a member function of a local class~~; substitution into the atomic constraints formed from them is instead performed as specified in [temp.constr.decl] and [temp.constr.atomic] when determining whether the constraints are satisfied or as specified in [temp.constr.decl] when comparing declarations.

Edit [temp.deduct.general]/8 as follows.

If a substitution results in an invalid type or expression, type deduction fails. An invalid type or expression is one that would be ill-formed, with a diagnostic required, if written in the same context using the substituted arguments.

[Note: ...]

Invalid types and expressions can result in a deduction failure only in the immediate context of the deduction substitution loci.

The *immediate context* of a construct *L* consists of the constructs that appear within *L*, subject to the following rules. Each

- body of a *lambda-expression* ([expr.prim.lambda]),
- construct that is considered separate from *L* for the purposes of template instantiation ([temp.decls.general]), regardless of whether it is instantiated at the same time as *L*

appearing within *L* is excluded from the immediate context of *L*. The following constructs are also included the immediate context of *L*:

- conversions applied to an expression or *braced-init-list* *E* that is in the immediate context of *L* in order to satisfy the requirements of the language construct in which *E* appears,
- implicitly generated destructor calls for temporary objects created within the immediate context of *L* ([class.temporary]),
- default template arguments used by *template-ids*, *simple-template-ids*, and *splice-specialization-specifiers* that are part of the immediate context of *L*, and
- those forming the immediate context of the *defining-type-id* of each alias template specialization ([temp.alias]) named by any construct that is in the immediate context of *L*.

[Note 6: The substitution into types and expressions can result in effects such as the instantiation of ~~class~~ template specializations ~~and/or function template specializations~~, the generation of implicitly-defined functions, etc. Such effects are not in the “immediate context” and can result in the program being ill-formed. —end note]

[Note: The immediate context defined above is unrelated to the context for class member access checking ([class.access]). —end note]

[Note: Default member initializers ([class.mem.general]) are not in the immediate context of the class object initialization construct. —end note]

Strike the normative text in [temp.deduct.general]/9 and the note, but leave the example (demonstrating the interaction of lambdas with SFINAE):

~~When substituting into a *lambda-expression*, substitution into its body is not in the immediate context.~~

~~[Note 7: The intent is to avoid requiring implementations to deal with substitution failure involving arbitrary statements.~~

~~[Example 7: ... —end example]~~

~~—end note~~

Demote the normative text in [dcl.attr.annotation]/4 to a note:

[Note: Substituting into an *annotation* is not in the immediate context. ~~—end note~~]

[*Example 1*: ...]

[*Example 2*: ...]

4. References

- [CWG2296] — Jens Maurer, "C++ Standard Core Language Active Issues, Revision 111", issue #2296, (https://www.open-std.org/jtc1/sc22/wg21/docs/cwg_active.html#2296).
- [N5032] — Thomas Köppe, "Working Draft, Standard for Programming Language C++" (<https://www.open-std.org/jtc1/sc22/wg21/docs/papers/2025/n5032.pdf>).
- [P0348R0] — Andrzej Krzemiński, "Validity testing issues" (<http://www.open-std.org/jtc1/sc22/wg21/docs/papers/2016/p0348r0.html>).
- [P2285R1] — Andrzej Krzemiński, Tomasz Kamiński, "Are default function arguments in the immediate context?" (<https://www.open-std.org/jtc1/sc22/wg21/docs/papers/2026/p2285r1.html>).
- [63391] — llvm/llvm-project GitHub issue, "clang++: Surprising SFINAE behavior for default function arguments" (<https://github.com/llvm/llvm-project/issues/63391>).